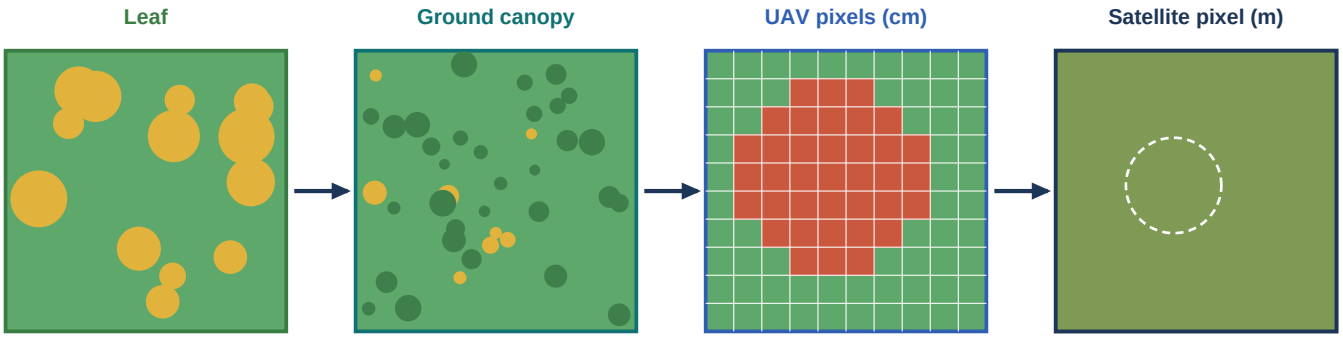
Leaf to Satellite: Crop Disease Detection Methodology

Concepts and an integrated, step-wise methodology for detecting crop disease at leaf, ground canopy, UAV and satellite level, and for linking the levels into one chain

Scope. This document covers **concepts and methodology only**. The methods of 12 published studies are merged into one workflow; each flow-chart box and table row names the studies that the step is based on (author, year). Full references are in Section 9.

1. Core concepts

- **Disease has a spectral signature.** Infection lowers chlorophyll (visible reflectance rises), damages leaf and canopy structure (NIR reflectance falls) and restricts water flow (canopy temperature rises). Each effect is read in a different spectral region (Liu Y. 2026; Poblete 2023; Yuan 2025).
- **The signal is measured at four levels.** Leaf (mm–cm), ground canopy (about 1 m²), UAV (1–10 cm pixels over fields) and satellite (0.5–20 m pixels over regions). Finer levels see the disease clearly but cover little ground; coarser levels cover large areas but mix healthy and diseased plants (Zhao 2023; Blasch 2023).
- **Upscaling means carrying knowledge up the levels.** Sensitive bands found on leaves are tested on canopies; canopy findings are converted into satellite bands; UAV maps provide pixel-sized labels for satellite models (Zheng 2018; Revill 2020; Jiang 2023).
- **Three links must match between levels:** the **bands** (spectral link), the **footprint** (spatial link) and the **calibration or model** (model link), with reflectance values made comparable (radiometric link) (Revill 2020; Yuan 2025; Poblete 2023).
- **Ground truth flows through every level.** Disease is scored in georeferenced 1 m² frames at the time of each measurement, so every level is calibrated and validated against the same plots (Zheng 2018; Guo 2021; Yuan 2025).



The same infection occupies a shrinking share of each measurement as the footprint grows, so the spectral contrast is diluted (Yuan 2025; Zhao 2023).

Figure 1. Concept of signal dilution with scale.

2. The four-level framework and its links

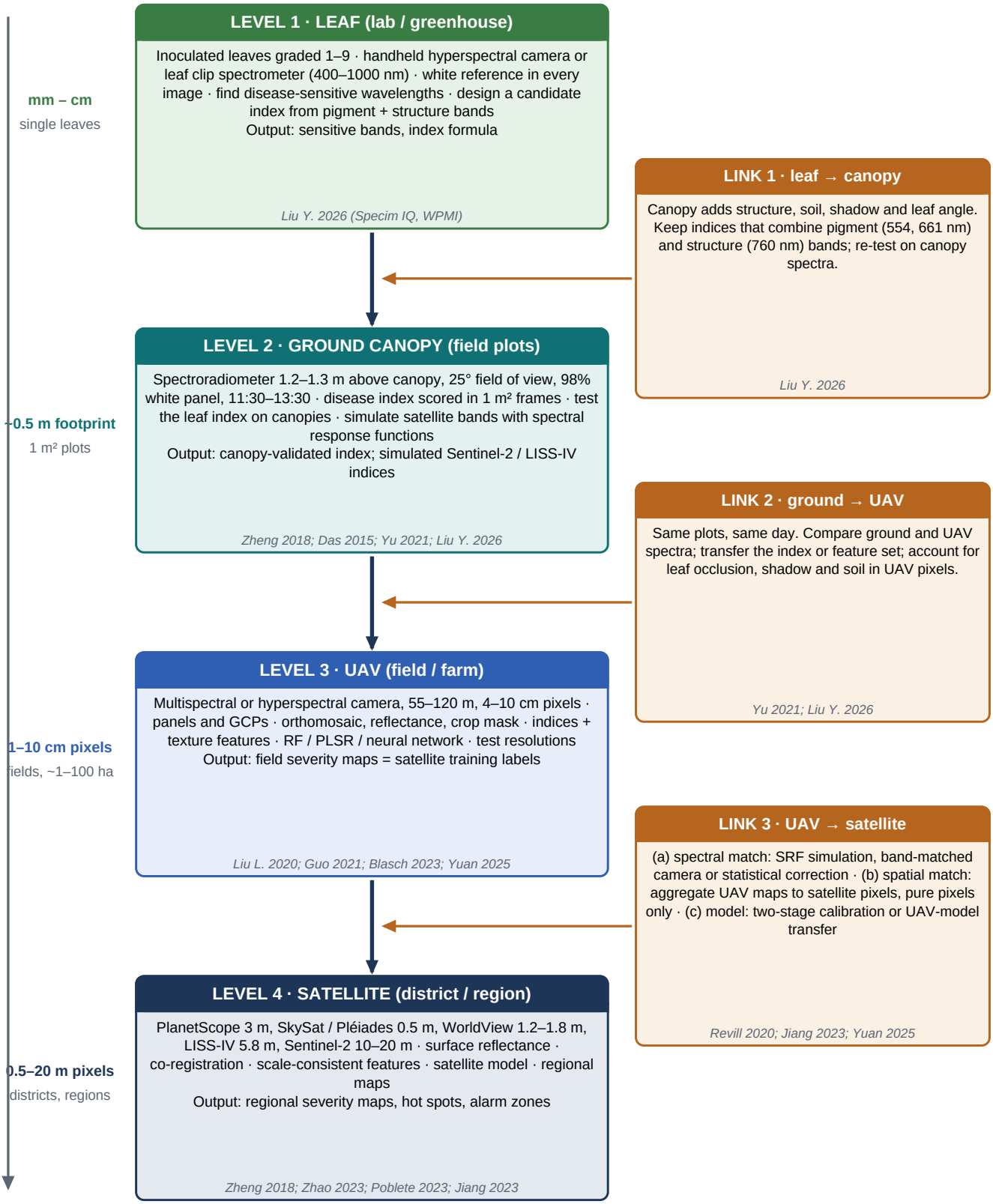


Figure 2. The four detection levels (left), the scale each works at (far left) and the method that links each level to the next (right). Arrows point in the upscaling direction.

3. Integrated step-wise methodology

All levels combined into one sequence of 16 steps. Colour shows the level of each step; amber steps are the links between levels.



Figure 3. Integrated workflow from leaf to satellite (S1–S16), with the studies each step is based on.

Step	What is done	How (method)	Based on
S1	Define symptoms	List the physiological effects of the pathogen and the spectral region each affects (visible, red edge, NIR, thermal)	Liu Y. 2026; Poblete 2023
S2	Design experiment	Inoculated, naturally infected and fungicide-protected plots; varieties of differing resistance; plot size larger than the coarsest pixel; 1 m² sample frames	Blasch 2023; Guo 2021; Liu Y. 2026
S3	Leaf spectra	Grade leaves 1–9 after inoculation; hyperspectral leaf images (400–1000 nm) with a white reference in each frame	Liu Y. 2026
S4	Sensitive bands and index	Single-band classification and band-combination search; build an index from pigment (green peak, red trough) and structure (NIR) bands	Liu Y. 2026; Zheng 2018
S5	Canopy spectra and scoring	Spectroradiometer at 1.2–1.3 m, 25° field of view, white panel, near solar noon; repeated readings averaged; disease index from 0–9 plant scores	Zheng 2018; Yu 2021; Liu Y. 2026
S6	Simulate satellite bands	Convolve canopy spectra with each sensor's spectral response function to obtain Sentinel-2, LISS-IV or other band values; re-derive indices	Zheng 2018; Das 2015

Step	What is done	How (method)	Based on
S7	UAV acquisition	Satellite-matched multispectral or hyperspectral camera; 55–120 m altitude; 75–85% overlap; solar noon ± 2 h; reflectance panels; GCPs or RTK	Revill 2020; Blasch 2023; Yuan 2025
S8	UAV processing	Structure-from-motion orthomosaic; radiometric calibration; crop/soil/shadow mask; inner plot buffer or tree-crown segmentation	Blasch 2023; Poblete 2023; Jiang 2023
S9	UAV features and model	Bands, indices and texture; feature selection; random forest, PLSR, SVM, KNN or neural network; resample to test pixel sizes	Liu L. 2020; Guo 2021; Zhao 2023
S10	Satellite acquisition	Cloud-free scene within a few days of the campaign; surface reflectance; pan-sharpening for VHR data	Zheng 2018; Blasch 2023; Poblete 2023
S11	Radiometric match	Co-register to GCPs or the UAV image; empirical-line cross-calibration or rescaling of satellite bands to UAV statistics	Yuan 2025; Poblete 2023; Zhao 2023
S12	Spatial match	Aggregate UAV maps to the satellite grid; use only pure in-field pixels away from edges and roads	Jiang 2023; Revill 2020
S13	Satellite model	Choose one: index thresholds, two-stage calibration (ground → UAV → satellite), or transfer of the UAV model using scale-consistent features	Zheng 2018; Revill 2020; Yuan 2025
S14	Map and track	Severity maps for each date; spatial hot-spot analysis to follow spread and recovery	Liu Y. 2026; Zheng 2018
S15	Validate	Independent plots or a second district; state the detection limit by growth stage and patch size	Jiang 2023; Zheng 2018; Zhao 2023
S16	Early warning	Where early stages matter, add narrow-band (PRI, SIF) and thermal (crop water stress) measurements from UAV or aircraft	Poblete 2023

4. Methodology at each level

4.1 Leaf level: finding the disease signal

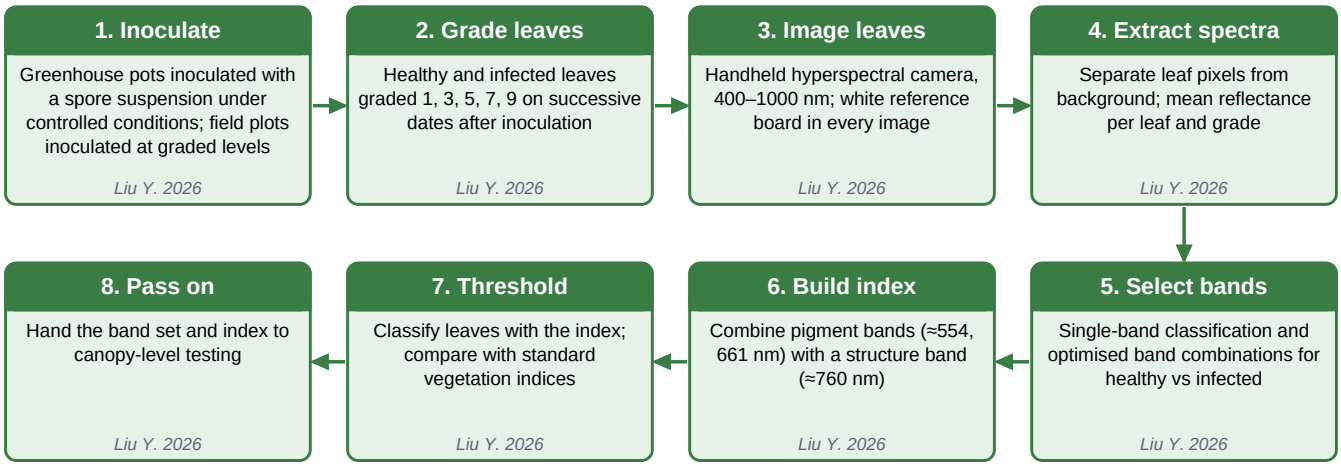


Figure 4. Leaf-level workflow.

4.2 Ground canopy level: confirming the signal in the field

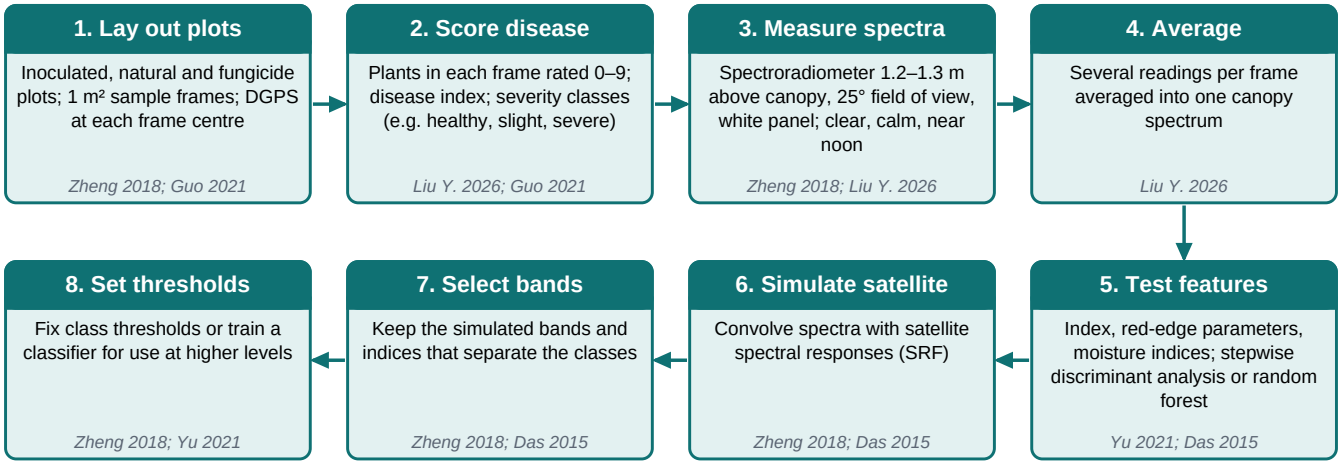


Figure 5. Ground canopy workflow.

4.3 UAV level: mapping disease across fields

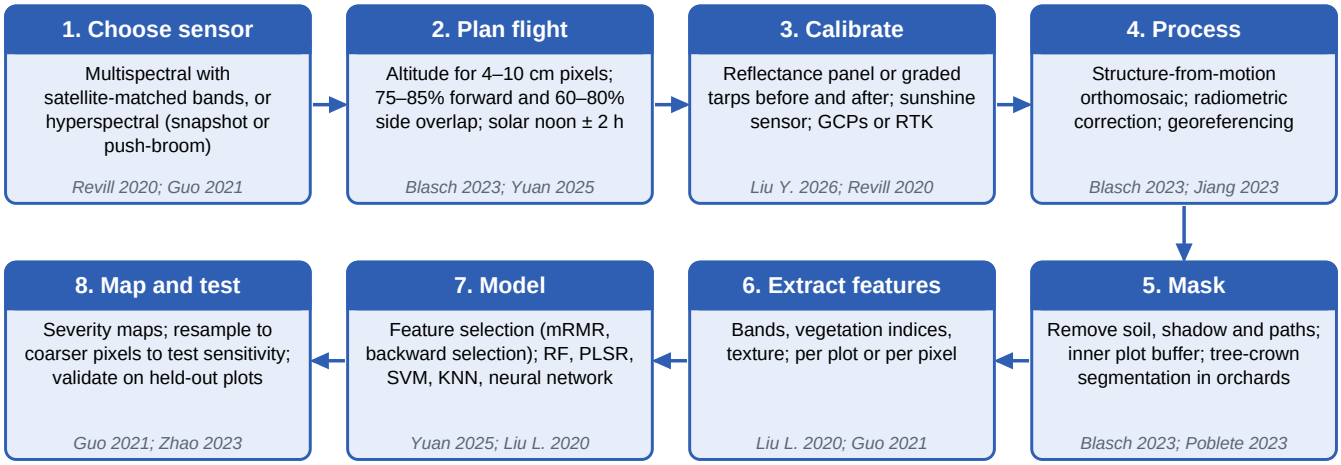


Figure 6. UAV workflow.

Study	Platform and camera	Flight / pixel	Calibration
Liu L. 2020	UAV + Cubert S185 snapshot hyperspectral	60 m, 4 cm; 80/65% overlap	Reflectance panel
Guo 2021	UAV + UHD 185 snapshot hyperspectral	1.2 cm, resampled to 3–20 cm	Reflectance panel
Yu 2021; Liu Y. 2026	DJI M600 + Resonon Pika L push-broom	120 m, ≈10 cm	White board; tarps 10/30/70%
Blasch 2023	Parrot Sequoia (G, R, red edge, NIR)	60 m, 5.6 cm; 80/80% overlap	Panel + sunshine sensor; 6 GCPs
Yuan 2025	DJI M600 + MicaSense RedEdge-M	55 m, 4 cm; 80/70% overlap	White panel
Revill 2020	DJI M600 + MAIA (Sentinel-2-matched bands)	100 m, 4.7 cm	Lambertian targets measured with ASD
Jiang 2023	eBee fixed-wing + Parrot Sequoia	100 m, 10 cm; 75% overlap	Pix4D calibration; GCPs
Zhao 2023	Hexacopter + twin Nikon D810 (RGB, NIR)	120 m, 2.9 cm; 85/60% overlap	Panels; 8 GCPs

4.4 Satellite level: regional mapping

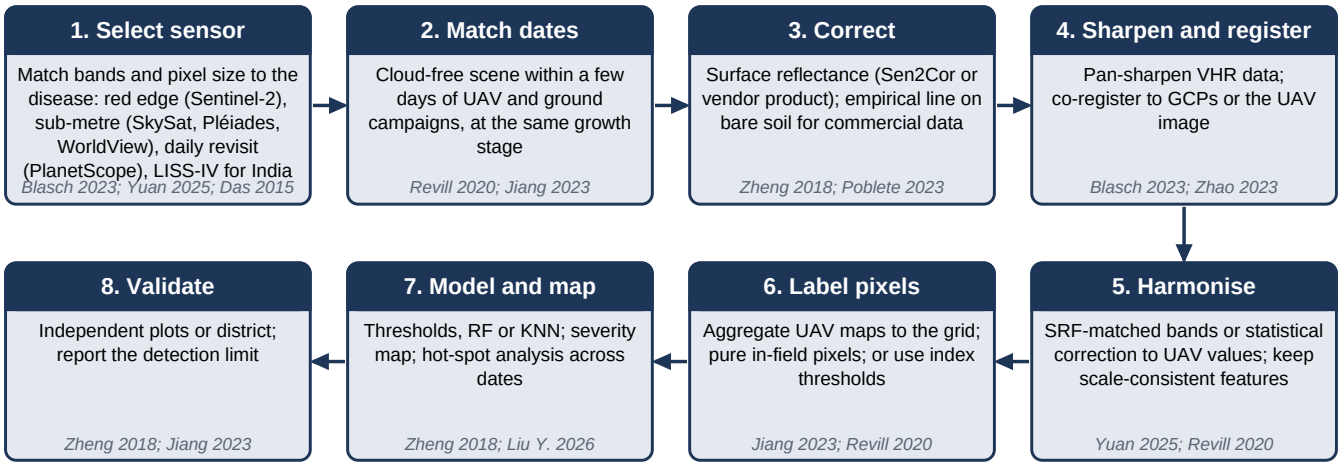


Figure 7. Satellite workflow.

5. Linking the levels

5.1 Spectral link: simulating satellite bands from spectra

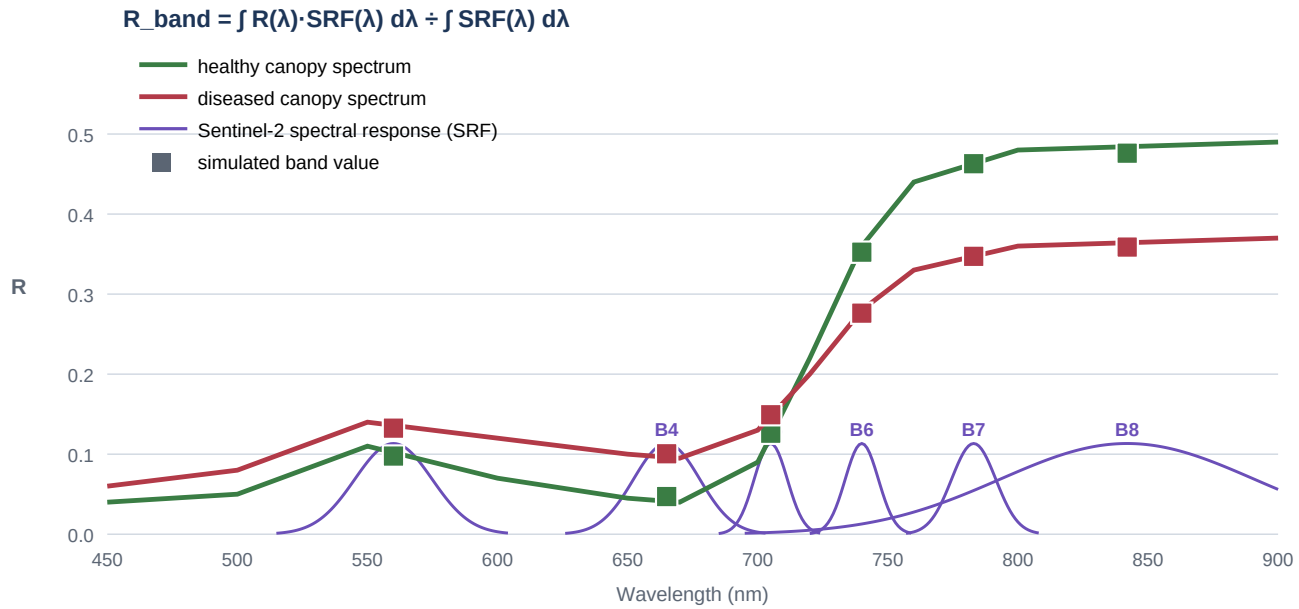
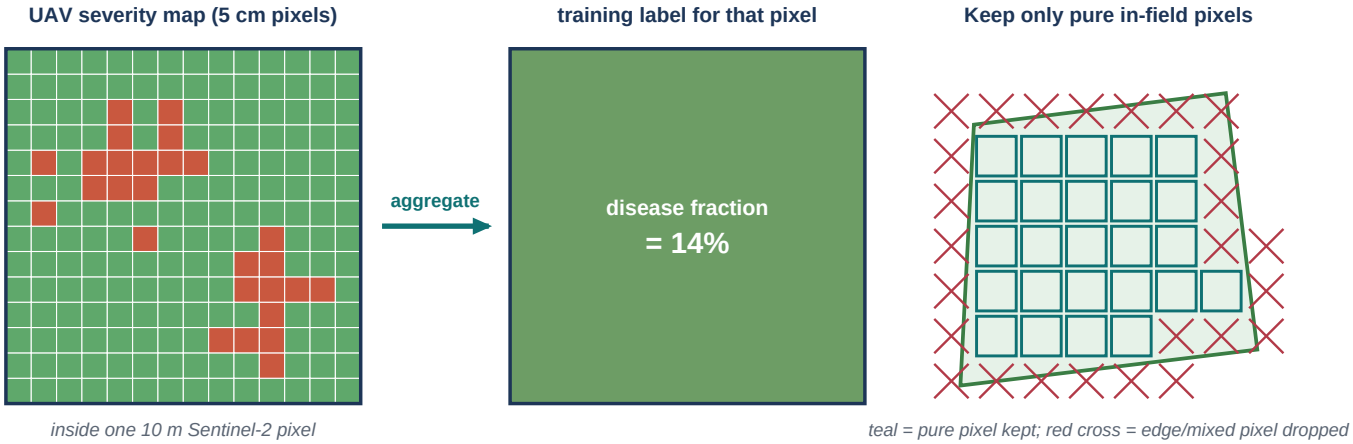


Figure 8. A canopy spectrum (line) is weighted by each band's spectral response (purple) to give the value that band would record (squares). Method of Zheng 2018 (Sentinel-2) and Das 2015 (LISS-IV). Spectra are illustrative.

- **Alternative 1:** a UAV camera whose bands copy the satellite's bands, which removes band-position differences at source (Revill 2020).
- **Alternative 2:** restrict analysis to bands every platform shares, e.g. green, red and NIR (Blasch 2023).

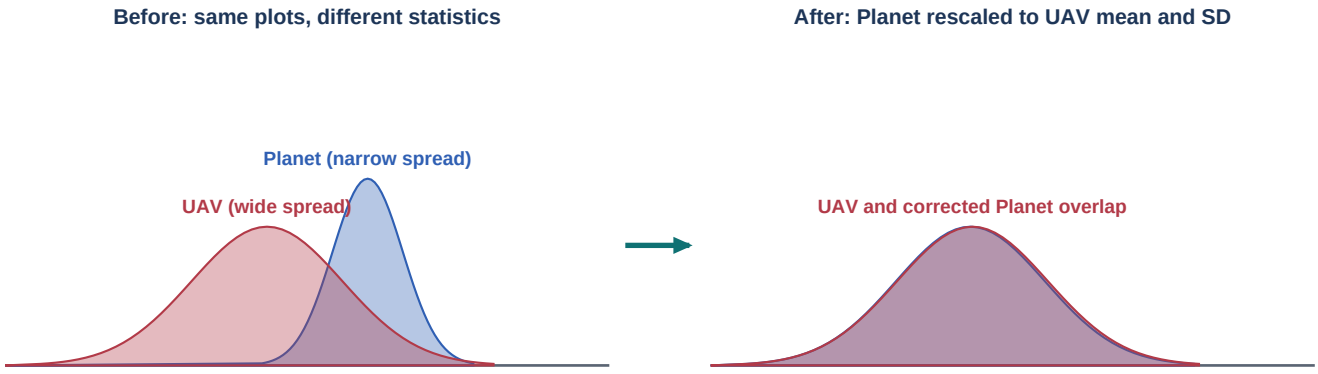
5.2 Spatial link: from UAV pixels to satellite pixels



Pixel aggregation and pure-pixel sampling (Jiang 2023; Revill 2020); 1 m plot buffers (Blasch 2023).

Figure 9. The UAV map inside one satellite pixel becomes one label (left); pixels crossing field edges or roads are dropped (right). Aggregation does not fully reproduce a real coarser sensor, whose optics also blur the image (Zhao 2023). Values are illustrative.

5.3 Radiometric link: making reflectance comparable



$$R' = (R_Planet - mean_Planet) \times (SD_UAV \div SD_Planet) + mean_UAV$$

Statistical correction (Yuan 2025). Physical alternative: empirical-line calibration on bare-soil targets against Sentinel-2 (Poblete 2023).

Figure 10. Satellite reflectance over the same plots has a narrower spread than UAV reflectance because each pixel averages healthy and diseased plants; rescaling aligns the two distributions so UAV-derived features can be applied (Yuan 2025). Schematic.

5.4 Model link: three strategies

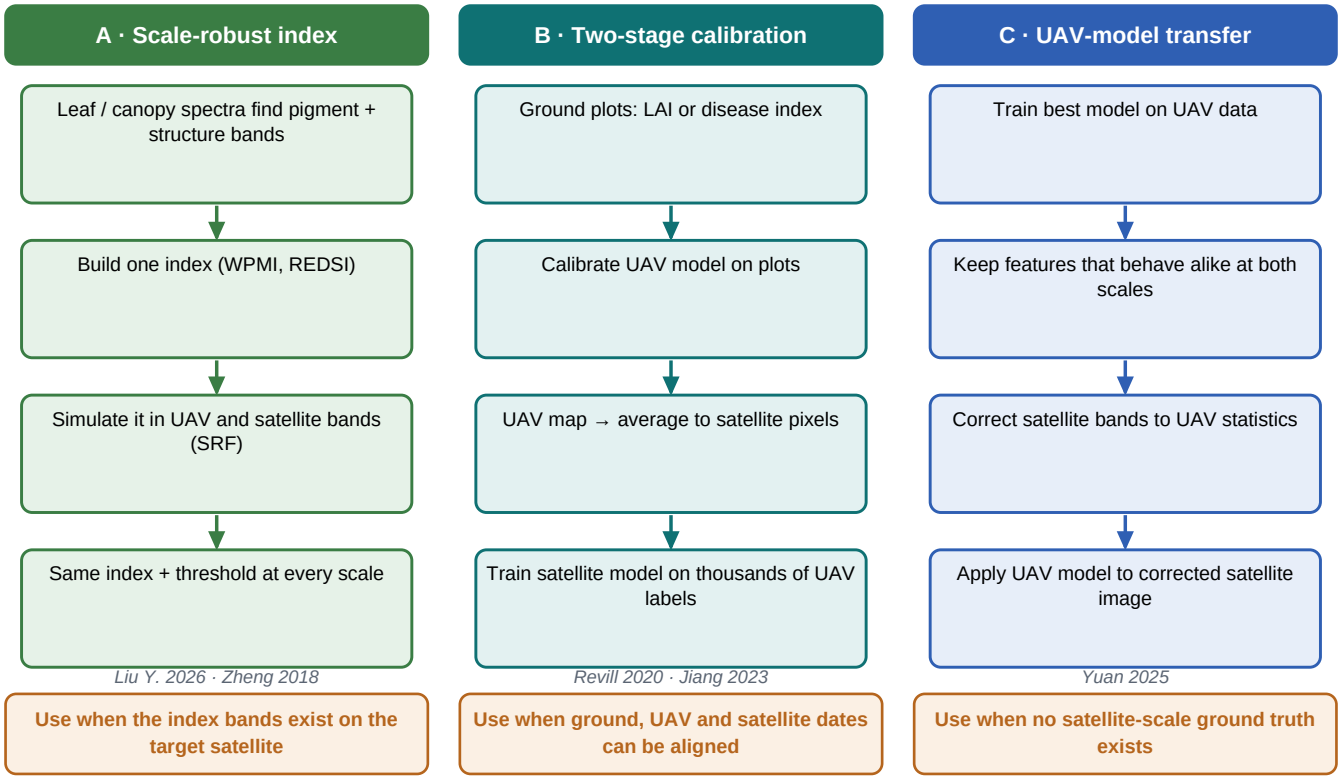
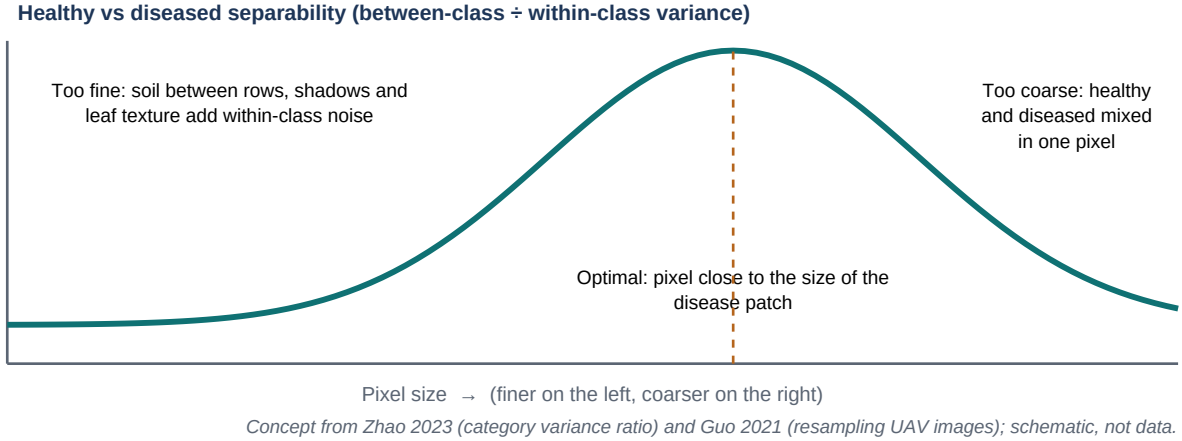


Figure 11. A: one index valid at all scales. B: the UAV converts a few ground points into many pixel-sized labels. C: a model trained on UAV data is applied to satellite data after spectral correction.

Strategy	Inputs needed	Concept	Caution
A · Scale-robust index	Leaf and canopy spectra; satellite SRFs	Physically based index combining pigment and structure bands, applied with the same logic at every level	Thresholds can shift with region, variety and stage
B · Two-stage calibration	Ground plots + UAV flights near satellite overpass	Ground → UAV model → UAV maps averaged to pixels → satellite model	Needs UAV campaigns each season
C · UAV-model transfer	UAV model; overlapping satellite scene	Align satellite bands to UAV statistics; keep features that behave alike; reuse the UAV model	Empirical correction; some detail lost at field edges

6. Scale effects to account for



Concept from Zhao 2023 (category variance ratio) and Guo 2021 (resampling UAV images); schematic, not data.

Figure 12. Concept of the optimal pixel size: separability between healthy and diseased crop rises as pixels coarsen from centimetres (less soil and row noise), peaks near the size of the disease patches, then falls as pixels mix classes.

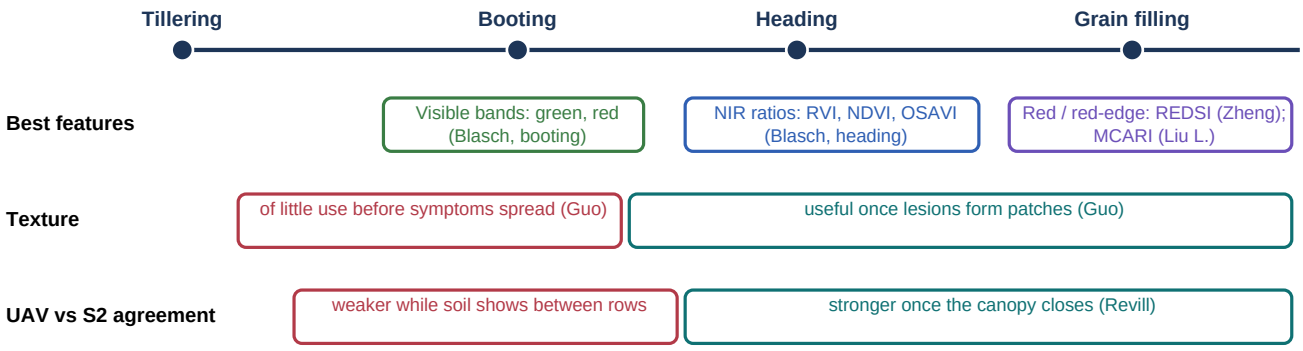


Figure 13. Useful features change with growth stage, and UAV and satellite signals agree better once the canopy closes (Blasch 2023; Guo 2021; Revill 2020; Zheng 2018; Liu L. 2020).

- **Early stages need fine spectral detail.** Broad satellite bands cannot measure traits such as PRI, SIF or anthocyanins that change first; use hyperspectral and thermal data for early detection (Poblete 2023).
- **Texture depends on resolution, spectral indices much less.** Satellite models should rely on spectral and temporal features (Guo 2021).
- **Pigment signals dominate at leaf scale, structure at canopy scale.** Indices combining both transfer best across levels (Liu Y. 2026).

7. Choosing the level for a task

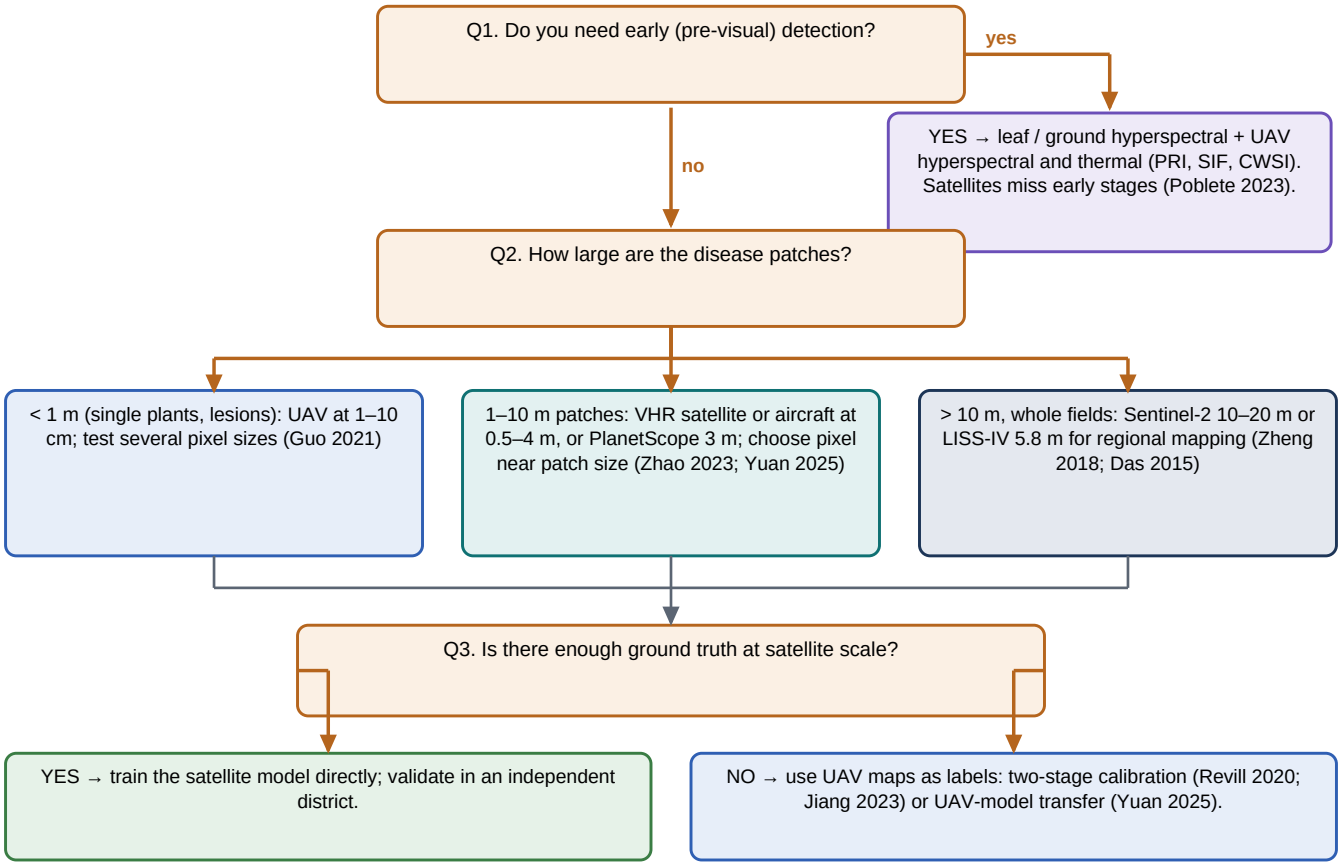


Figure 14. Decision chart linking the detection goal and patch size to the platform, and the availability of ground truth to the model-link strategy.

8. Example chain: wheat yellow rust

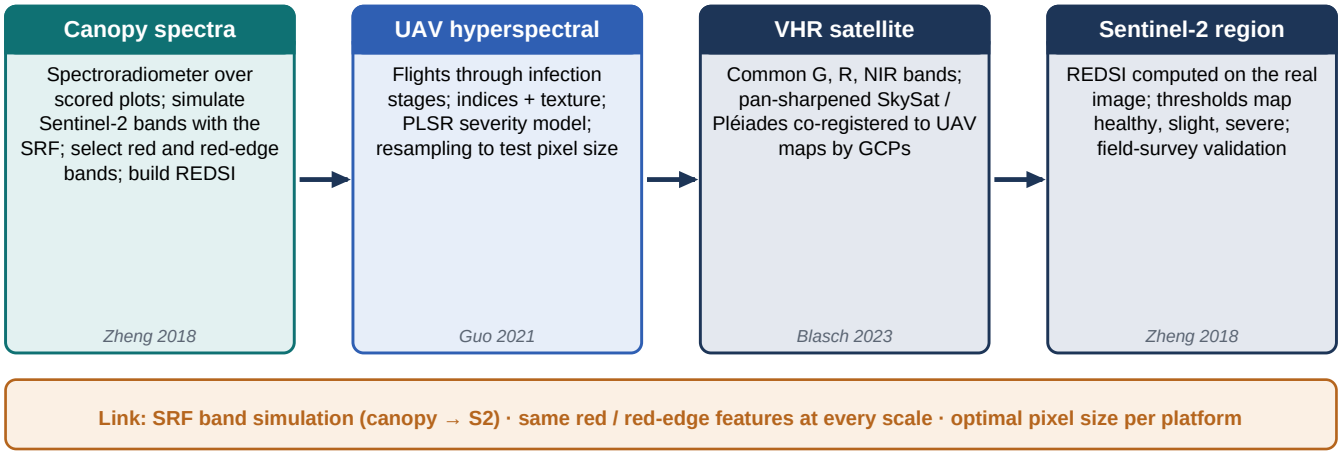


Figure 15. How the methods of three studies chain together for one disease: canopy spectra to Sentinel-2 (Zheng 2018), UAV hyperspectral monitoring (Guo 2021) and UAV-to-VHR transfer (Blasch 2023).

8.1 Key equations

Quantity	Equation	Based on
Satellite band from spectra	$R_{band} = \int R(\lambda) \cdot SRF(\lambda)\, d\lambda \div \int SRF(\lambda)\, d\lambda$	Zheng 2018; Das 2015
Red Edge Disease Stress Index	$REDSI = [(705 - 665)(R_{B7} - R_{B4}) - (783 - 665)(R_{B5} - R_{B4})] \div (2 R_{B4})$	Zheng 2018
Wheat powdery mildew index	$WPMI_G = (R760 - R554)/(R661 + R554)$; $WPMI_R = (R760 - R661)/(R661 + R554)$	Liu Y. 2026
Coefficient of infection (rust)	CI = severity (%) × host-response constant	Blasch 2023
Spectral correction	$R' = (R - mean_{sat}) \times (SD_{UAV} \div SD_{sat}) + mean_{UAV}$	Yuan 2025
Separability for pixel-size choice	CVR = between-class variance ÷ within-class variance	Zhao 2023
Pixel label from UAV map	f = diseased UAV pixels ÷ all crop UAV pixels inside the satellite pixel	Jiang 2023; Revill 2020

9. References

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Zheng, Q., Huang, W., Cui, X., Shi, Y. & Liu, L. (2018). New spectral index for detecting wheat yellow rust using Sentinel-2 multispectral imagery. *Sensors*, 18(3), 868. https://doi.org/10.3390/s18030868

Revill 2020 and Jiang 2023 study LAI and nitrogen, not disease; their scaling methods are used here because they transfer directly. Das 2015 details are from the published abstract.